

TechTwin AI Teaching Assistant

Complete Data Dictionary

1. Role Table

Field Name	Data Type	Description
role_id	INT (PK)	Unique Role ID
role_name	VARCHAR(30)	Student/Teacher/Admin/HOD/SuperAdmin
description	VARCHAR(200)	Role responsibilities

Purpose: Stores the five main system roles.

2. User Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique User ID
name	VARCHAR(100)	Full Name
email	VARCHAR(100)	Login Email
password	VARCHAR(255)	Encrypted Password
role_id	INT (FK)	References Role table
status	BOOLEAN	Active/Inactive
created_at	DATETIME	Account creation date

Purpose: Stores login and identity details of all users.

3. Question Table

Field Name	Data Type	Description
question_id	INT (PK)	Unique ID
student_id	INT (FK)	References User
subject	VARCHAR(100)	Selected subject
question_text	TEXT	Student question
asked_at	DATETIME	Date & Time asked

Purpose: Stores questions asked by students.

4. Answer Table

Field Name	Data Type	Description
answer_id	INT (PK)	Unique ID
question_id	INT (FK)	References Question
answer_text	TEXT	AI generated answer
generated_at	DATETIME	Time generated

Purpose: Stores AI responses for student questions.

5. StudyMaterial Table

Field Name	Data Type	Description
material_id	INT (PK)	Unique ID
teacher_id	INT (FK)	References User
subject	VARCHAR(100)	Subject name
file_name	VARCHAR(255)	Uploaded file name
upload_date	DATE	Upload date

Purpose: Stores study materials uploaded by teachers.

6. Course Table

Field Name	Data Type	Description
course_id	INT (PK)	Unique ID
course_name	VARCHAR(100)	Course title
status	VARCHAR(20)	Pending/Approved
created_by	INT (FK)	Admin ID
approved_by	INT (FK)	HOD ID

Purpose: Manages and approves academic courses.

7. Report Table

Field Name	Data Type	Description
report_id	INT (PK)	Unique ID
report_type	VARCHAR(50)	Performance/Activity
generated_by	INT (FK)	References User
generated_date	DATE	Report date

Purpose: Stores system and academic reports.

8. StudentActivity Table

Field Name	Data Type	Description
activity_id	INT (PK)	Unique ID
student_id	INT (FK)	References User
activity_type	VARCHAR(100)	Login/Ask Question
activity_time	DATETIME	Timestamp

Purpose: Tracks student activity for monitoring.

9. AdminManagement Table

Field Name	Data Type	Description
admin_id	INT (FK)	References User
managed_by	INT (FK)	SuperAdmin ID
updated_at	DATETIME	Last updated time

Purpose: Controls and manages Admin accounts.